

RAKESH ALGOT

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PYTHON DEVELOPER | GENERATIVE AI & FASTAPI

Python Developer with experience building scalable FastAPI applications, AI-powered retrieval systems, asynchronous processing pipelines, and production-ready Generative AI applications. Skilled in developing REST APIs, integrating LLMs, designing RAG pipelines, implementing vector search, and deploying cloud-native applications. Experienced in building end-to-end AI workflows involving document processing, retrieval optimization, prompt engineering, and AI service integrations.

CORE SKILLS

Language & Frameworks: Python, FastAPI, REST APIs.

Databases: MongoDB, PostgreSQL, Redis.

Generative AI: LLM Integration, Retrieval-Augmented Generation (RAG), LangChain, Qdrant, FAISS

Cloud & DevOps: Docker, AWS EC2, Linux, Nginx, GitHub Actions.

Authentication & Security: RBAC, Keycloak, Authentication, Authorization

Version Control: Git, GitHub

Testing & API Tools: Pytest, Postman, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Python Developer

Yensi Solutions — Hyderabad | Aug 2024 – Present

- Reduced API response latency by ~40% through Redis caching, query optimization, and efficient data retrieval strategies.
- Designed asynchronous document-ingestion pipelines using FastAPI Background Tasks, Celery, Redis, and MongoDB for large-scale document processing.
- Designed, developed, and maintained 100+ production-grade FastAPI endpoints supporting document processing, user management, authentication, AI workflows, and business operations.
- Built retrieval workflows processing 300+ documents using Qdrant vector search, hybrid retrieval, and reranking techniques to improve retrieval quality and response relevance.
- Improved retrieval accuracy and response quality by optimizing hybrid search, ranking, and reranking pipelines, achieving ~20–30% improvement.
- Deployed and managed containerized applications on AWS EC2 using Docker, Nginx, and Linux environments.

PROJECTS

Citex — Citation-grade Context Retrieval System

Tech: FastAPI, MongoDB, Qdrant, Redis, Docling, BM25, Celery

- Built asynchronous document ingestion pipelines for large-scale PDF and document processing.
- Developed hybrid retrieval systems combining semantic vector search, BM25 keyword search, and reranking.
- Implemented document chunking, metadata extraction, and citation mapping for accurate evidence-backed responses.
- Improved retrieval latency through Redis caching and retrieval pipeline optimizations.

AH Ledger — AI-Powered S106 Compliance System

Tech: FastAPI, MongoDB, Claude, Gemini, OCR, AsyncIO

- Developed AI-driven document processing pipelines for legal clause extraction and compliance tracking.
- Implemented prompt engineering, validation pipelines, fallback handling, and hallucination reduction mechanisms.
- Built OCR-enabled processing workflows for scanned and image-based documents.
- Developed document comparison engines using SequenceMatcher and conflict detection logic.

AI Pujari — AI-Enabled Digital Platform

Tech: FastAPI, MongoDB, ElevenLabs (STS/TTS), HeyGen API, FFmpeg, Keycloak, Razorpay

- Integrated ElevenLabs and HeyGen APIs to generate personalized AI voice and video experiences.
- Built asynchronous AI processing workflows for long-running media generation tasks.
- Developed payment gateway integrations with Razorpay and Stripe, including dynamic currency routing, Developed secure authentication and role-based access control using Keycloak.

Golf Addicts — Tournament & League Platform

Tech: FastAPI, MongoDB, Redis, Docker, Keycloak

- Developed FastAPI-based REST APIs for player management, tournament setup, match scheduling, and leaderboard generation.
- Implemented CSV import/export workflows with validation, duplicate detection, and error handling.
- Designed caching layers using Redis and implemented MongoDB indexing to accelerate data retrieval, improve scalability, and optimize high-traffic API performance.
- Integrated Keycloak-based authentication and role-based authorization for secure API access.

EDUCATION

Master of Computer Applications (MCA)

Nizam College — Hyderabad

Graduated - 2022